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ORIGINAL ARTICLE

Machine learning approaches for real-time process anomaly detection in wire arc additive manufacturing

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Abstract
In gas metal arc welding (GMAW) processes, including wire arc additive manufacturing (WAAM), machine learning (ML) is emerging as a powerful tool for monitoring both process and product anomalies. However, a significant challenge in real industrial environments is the reliance on large, balanced datasets for training supervised learning models. To address this issue, a shift toward unsupervised learning is gaining attention in this research field, offering the potential to work effectively with small and unbalanced datasets. However, different materials, sensors, and welding technologies have been used in the literature, making complex the comparison of the results. This work fills that gap by presenting a comprehensive comparison of both supervised and unsupervised learning methods. An experimental campaign was conducted on Invar 36 alloy—a material with limited WAAM research—where 15 wall structures were deposited with varying process parameters using the natural dip transfer process, aiming to identify the optimal parameters for this alloy. Data on welding current and voltage were captured, and during the qualification procedure, anomalies were detected, some of which led to product defects. Supervised, unsupervised, and semi-supervised ML approaches, along with a detailed frequency domain analysis of the collected signals, were applied to process the obtained unbalanced dataset. The results provide key insights: while supervised learning models can be applied to anomaly detection in small and unbalanced datasets, they are prone to overfitting, which limits their practical use due to the prevalence of normal cases over anomalies in the dataset, resulting in higher number of missed anomalies. In contrast, unsupervised models, with their lower generalization capability, tend to exhibit higher false alarm rates but better performance to identify anomalous data. This work not only compares in depth these data analytics methodologies but also offers guidance on selecting the appropriate ML algorithm based on specific industrial objectives and provides insights into the printability of Invar 36 for WAAM applications under natural dip transfer process.

Keywords Wire arc additive manufacturing · Monitoring · Supervised learning · Unsupervised learning · Invar 36 · Machine learning

1 Introduction

1.1 The problem of anomaly detection for small and unbalanced dataset

Anomaly detection involves identifying patterns in data that significantly deviate from a predefined normal behaviour [1], and it plays a crucial role in ensuring quality and safety across various industrial processes. Today, the capability for anomaly detection has become important to many fabrication processes, including machining [2], welding [3], additive manufacturing (AM) [4], and forming [5]. As industry interest in anomaly detection grows, numerous studies have been conducted on the topic [6–11].

The analysis of those works highlights that traditional machine learning (ML)–based methods for anomaly detection often depend on large, balanced datasets to train supervised models, in which the problem is treated as a binary classification. However, the challenge lies in developing methods that can function effectively with smaller or unbalanced datasets, which are typical in industrial settings. This is because failures are inherently rare, resulting in limited data on such anomalies, while data from normal operating conditions is more readily available.

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Therefore, in many practical scenarios, obtaining such balanced and labelled datasets can be challenging and expensive. This is particularly true in specialized fields like welding, where the cost of producing a large volume of both high-quality and low-quality or failures data, e.g. related to cracks, porosity, and lack of fusion, can be prohibitive. When dealing with small, unbalanced datasets, the task of detecting anomalies becomes even more complex.

Given this context, effectively and cost-efficiently detecting anomalies in situations where only limited and potentially unbalanced data is available remains a significant challenge, especially in arc welding processes. This issue is underrepresented in the literature, with few proposed solutions addressing it comprehensively.

1.2 Wire arc additive manufacturing

Wire arc additive manufacturing (WAAM) [12, 13], illustrated in Fig. 1, is a direct energy deposition (DED) process that uses arc welding technology [14] to build large metal components layer by layer. Since its development in the early 90 s [15, 16], WAAM has advanced with innovations in welding, automation, and ICT, making it more sustainable and effective for producing defect-reduced, high-strength parts. Widely applied across industries like aerospace and construction, WAAM often utilizes gas metal arc welding (GMAW) for its cost-effectiveness and automation ease. However, WAAM can introduce defects due to process instability, influenced by parameters like contact tip to workpiece distance (CTWD) and arc length [17–19]. This makes process monitoring crucial for adjusting the process in real-time and assessing quality [20–22]. Controlled dip transfer methods, such as cold metal transfer [23] or surface tension transfer [24, 25], are now commonly used to stabilize the process, reducing defects and improving mechanical properties and surface quality. These processes are typically used to print special alloys like Invar 36 and others [26–28], which presents challenging deposition characteristics due to its thermomechanical properties [29, 30]. However, also constant voltage arc welding processes can be used to print this alloy [31], and natural dip transfer represents a solution to mitigate the effects of the high heat input supplied to the material. In particular, concerning previous studies on printing Invar 36 using WAAM, Veiga et al. [32] utilized a GMAW spray transfer technique, whereas Sood et al. [33] explored the alloy's printability using various GMAW metal transfer modes. Their findings indicated that reducing heat input—characteristic of the dip transfer process—reduced the susceptibility of the material to ductility dip cracking. Additionally, Jiao et al. [34] investigated the CMT process, while Fowler et al. [30] applied the STT process, which helps minimize heat input and overall energy consumption.

However, there is currently limited information on how to print this alloy using the natural dip transfer process. Furthermore, the uncontrolled nature of droplet formation presents significant challenges for process monitoring applications aimed at detecting anomalies. This is because both the deposition process and short circuits occur in an unpredictable manner, making it more difficult to identify when a waveform deviates from the expected distribution, as seen in controlled processes.

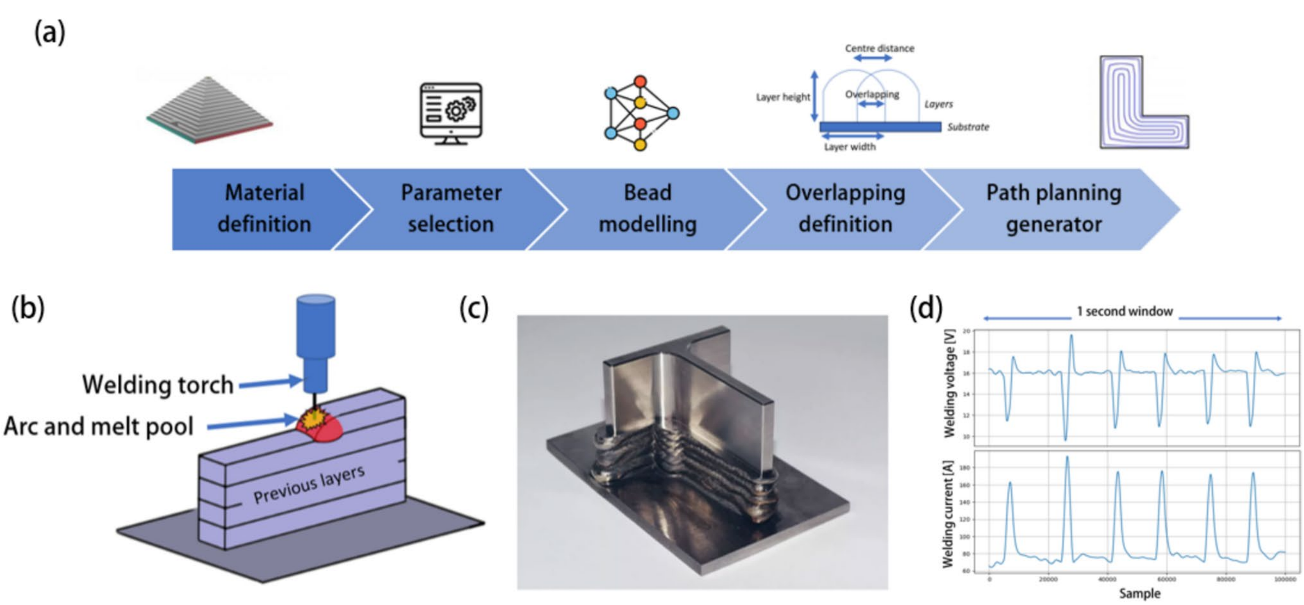


Fig. 1 **a** Wire arc additive manufacturing process use the **b** welding technology to deposit in a layer-by-layer fashion metal components **c** obtaining as result a near net shape geometry. **d** The natural dip transfer can be used as fundamental technology to reduce the heat input to the material

1.3 Process monitoring in gas metal arc welding processes: a review

In the field of arc welding and WAAM, a small dataset is often the result of rigorous procedure qualification processes [35, 36]. Welding procedure qualification, as illustrated in Fig. 2, involves testing and validating process parameters to ensure they meet specific quality and performance standards. This process typically generates a limited amount of data due to the high costs associated with producing and testing welds, as well as the time required for each qualification. Once the qualified process parameters are found, these parameters are used during the deposition process and the process itself is monitored against what was observed during qualification.

At this stage, it is important to distinguish between a *process anomaly* and a *product anomaly*, though both are types of irregularities. A process anomaly occurs in real time, involving issues with parameters like heat or arc stability. Meanwhile, a product anomaly is a defect in the final weld resulting from process anomalies or material issues. While process anomalies can signal potential issues, they do not always mean a defect has formed; they simply act as alerts that something in the process may need attention. On other hand, a defect detection software is able to detect defects within components. Regarding process monitoring, several studies have been conducted using different data analytics techniques and sensors such as welding current, welding voltage, acoustic emission, and images from the melting pool. For data analytics, both time-domain and frequency-domain analyses were conducted to extract useful features from the collected signals, along with image processing techniques applied to time–frequency representations of time domain signals and melt pool images.

1.3.1 Process monitoring using welding current and welding voltage sensors

Regarding process monitoring using welding voltage and welding current sensors, both process anomaly and product anomaly software have been developed from different authors for both GMAW and WAAM processes.

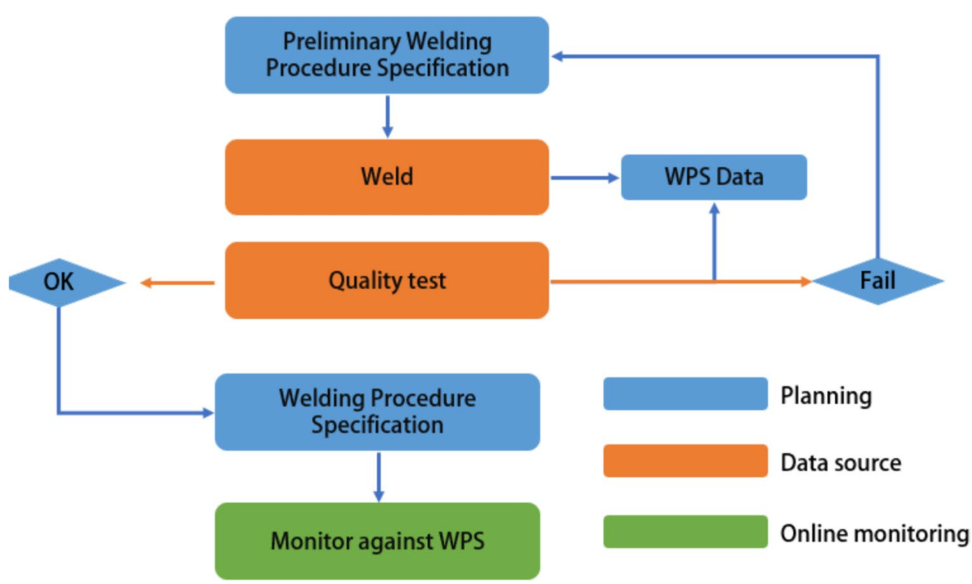
Regarding GMAW process, Nele et al. [37] and Shin et al. [38] proposed artificial neural networks (ANN) to detect online defects within components with an accuracy around 90%. Shin et al. extracted time-domain features from time-series signals of dip-transfer welding current and voltage, whereas Nele et al. used RMS values as inputs for an ANN.

Another technique for assessing weld quality after deposition is proposed by Sumesh et al. [39]. They employed a simple decision tree classifier to assess defects such as burn-through and porosity in GMAW of carbon steel with an accuracy of 90% using the features extracted in the time-domain. Moreover, Melakhsou and Batton-Hubert [40] used a non-parametric kernel classifier to detect burn-through defects. They proposed a methodology that processes the probability density distribution (PPD) obtained from the time-domain voltage signal during mild steel welding.

Also in the field of WAAM, authors proposed the employment of supervised learning for defect detection. Zhang et al. [41] employed time–frequency domain analysis and convolutional neural networks (CNN) to detect surface oxidation defects at the end of the deposition of NAB alloy components.

Regarding online process anomaly detection, Li et al. [42] proposed a methodology based on support vector machine to process the extracted time-domain features aiming to classify process anomaly such as pool shift in a CMT-based of ER706-S mid-steel. The online nature of the proposed

Fig. 2 Illustration of how it is obtained and then used a Welding Procedure Specification



methodology allows to pinpoint the portion of the space where the anomaly happen, which is useful for further analysis.

As mentioned earlier, techniques that rely on supervised learning require a balanced dataset to achieve good performance, especially those based on deep learning. As a result, there has been a growing trend toward the use of unsupervised learning, particularly for process anomaly detection. A process anomaly may or may not be related to defects during production, but identifying its presence is crucial for non-destructive analysis. This is because non-destructive evaluation targets can be generated at specific points where an anomalous condition occurs [25].

Our research group has focused on this aspect in several papers, where various methodologies have been developed and compared with the aim of detecting process anomalies. These studies are briefly summarized, representing the current state of the art in this field.

In [43], an online unsupervised learning agent uses a digital twin to forecast the welding current and welding voltage values with a sample rate of 10 Hz during a natural dip transfer process of mild-steel. The proposed algorithm detects out-of-distribution values based on the prediction error. Anomaly in sensor signals highlighted the presence of porosity, humping, and burn through defects in the deposited layers. In [44], the certified process parameters were employed to collect data in a pulsed-GMAW process of Inconel 718, and the features extracted in both time and frequency domain have been used in deep learning algorithm to learn the hidden pattern of normality. Then, anomaly such as pool shift and porosity has been detected as deviation from the learnt pattern. Finally, in [45], several unsupervised techniques have been compared to evaluate their performance in process anomaly detection, highlighting the importance of frequency analysis of welding current and voltage signals to enhance the differentiation between signals compared to time-domain features. This analysis, conducted for mild steel printed using the STT process, emphasizes the presence of defects, such as porosity when process anomalies are identified.

1.3.2 Process monitoring using audio signals and weld pool images

In addition to welding current and welding voltage signals, also audio collected with microphone or acoustic emissions sensors can be used for this task, in which frequency domain analysis plays an important role. This is because sound is the primary sense used by experienced welders to assess the weld's condition and to manually adjust parameters, as it is also related to the mode of metal transfer. Regarding welding process, Yang et al. [46] applied time–frequency domain analysis to generate spectrograms of CMT welding on aluminium and demonstrated that CNNs are more effective than SVMs for detecting defects such as burn-through and lack of fusion. Jang et al. [47] used FFT to extract features from audio signal during a P-GMAW process and then compared the ability to detect defects such as porosity within welds using different ML algorithms such as ANN with an accuracy of 95%. Zhang et al. [48] used a similar approach for WAAM of high-manganese alloy comparing the results of different CNN architecture. They were able to detect defects such as discontinuity and porosity with a ResNet-18 architecture with 98%. Always in the field of WAAM, Surovi and Soh [49] used audio signals and ML to identify geometry defects, such as arc ignition and breakthrough which typically happen at the beginning and at the end of deposition process. Finally, the works of Rohe et al. [50] and Gourishetti et al. [51] demonstrated the ability of arc sounds to detect instabilities due to poor shielding gas or the presence of contaminants, which may results in defects like porosity. In this case, these authors proposed the processing of the spectrogram image obtained by time–frequency process of this audio signals trough CNN. Also in the field of audio monitoring, unsupervised learning is an emergent topic, as suggested by the work of Rahman et al. [52], which apply clustering to monitor the process by statistical features like the skewness of the FFT response. However, in this case, the possibility to monitor process anomaly with these sensors with this approach is still underexplored and requires more studies.

For what concern welding pool images, these data are the mostly used in process monitoring. Although they have been largely employed in the past for the development of feedback control strategies [53–55], in the last decade, the number of works related to process monitoring increased thanks to the democratization of deep learning [56].

Xia et al. [57] used the images collected online from the welding camera to detect several defects like misalignment, undercut, burn trough, and incomplete penetration during GMAW of stainless steel. They used a ResNet18 to reach an average accuracy among the classes of 99%.

Xia et al. in another work [58] proposed the usage of welding camera for process monitoring, being able to classify process anomaly such as spatter, which may turn into porosity defects. Also in this field, recent work has shifted the process monitoring paradigm toward unsupervised learning, for the reasons previously mentioned. Song et al. [59] proposed an algorithm based on convolutional autoencoder (CAE) which is able to identify anomalous condition of the welding pool which suggests process instability. Table 1 summarized the result of the conducted literature review analysis.

1.4 Proposal of this work

As discussed above, in arc welding-based processes, most online anomaly detection applications have been developed using supervised learning techniques. Moreover, these methodologies have primarily been applied to common materials and waveform welding methods, such as steel with P-GMAW, CMT, or STT. However, a comparative analysis of different methodologies for these processes is lacking, as existing studies often propose their own methods without an extensive comprehensive comparison. In this work, we aim to compare various ML methodologies, both supervised and unsupervised, to develop a process anomaly detection system using data collected from welding current and welding voltage sensors. The goal is to quantitatively demonstrate the effectiveness of unsupervised learning, particularly when working with unbalanced datasets as those typical of qualification procedure.

To illustrate the challenges and solutions in process anomaly detection with small datasets, we use a case study focused on the welding of Invar 36 using the natural dip transfer GMAW process. This case study presents several important insights. First, it allows us to discuss the application of anomaly detection to a special alloy that is less commonly found in the literature. This alloy has potential industrial applications; however, few studies have explored its printability, and none has investigated its printability under standard short-circuit mode. This mode enables a reduction in heat input and represents the natural form of feedback-controlled processes. With this study, we provide insights into WAAM of Invar 36 in the short-circuit mode. This study provides insights into WAAM of Invar 36 in the short-circuit mode, contrasting it with known processes such as spray transfer [31] and STT [30], which poses specific challenges due to the unique properties of the material and stringent quality standards.

Second, the use of a natural dip transfer process introduces challenges not present in waveform welding, such as the stochastic nature of the arc behaviour due to the lack of feedback control typical of waveform processes. This makes feature extraction techniques more difficult and less explored and more difficult to develop robust anomaly detection application.

2 Materials and methods

2.1 Supervised learning in small dataset

As reported in the review section, supervised learning models have been widely applied to both process and product anomaly detection tasks, typically framing the problem as a binary classification. Neural networks are the most used models for this purpose. However, other methods, such as isolation forests, decision trees, and logistic regression, can also be effectively employed for anomaly detection in arc welding-based processes.

These models typically require large, labelled datasets to be trained effectively. However, when dealing with

Table 1 Summary of process monitoring methodologies in arc welding-based processes

References	Sensor used	Methodology
[37, 38]	Welding current and welding voltage	Online time domain features extraction and ANN to solve weld process quality as a binary classification problem
[39]	Welding current and welding voltage	Post weld defect detection based on time domain features and decision-tree algorithm
[40]	Welding voltage	Post weld quality assessment based non-parametric kernel classifier using the time-domain PPD
[41]	Welding voltage	Post WAAM deposition to detect surface defects based on time-domain frequency analysis and CNN
[42]	Welding current and welding voltage	Online process monitoring using time-domain features and SVM classifier
[43]	Welding current and welding voltage	Use unsupervised GMM to detect online WAAM process anomaly based on digital twin-based estimation error
[44]	Welding current and welding voltage	Online time domain features extraction and time–frequency images processing via double autoencoder for WAAM process anomaly detection
[45]	Welding current and welding voltage	Online WAAM process anomaly detection based on time-domain frequency analysis and CNN
[46–48, 50, 51]	Microphone	Post weld defect detection using time–frequency domain spectrogram analysis and CNN
[49]	Microphone	Post deposition quality assessment based on statistical features extraction from audio signal in both time and frequency domain and SVM
[52]	Acoustic emission	Statistical features extraction from acoustic emission signal and clustering via K-Means for process monitoring
[57, 58]	Welding camera	Using CNN for online process anomaly or defect detection
[59]	Welding camera	Using autoencoder for online process anomaly using unsupervised learning paradigm

small datasets, several critical challenges arise that can impact the performance and reliability of anomaly detection systems. Firstly, small datasets often lack sufficient number of examples, which leads to a scarcity of representative samples. This limitation hinders the ability of supervised learning algorithms to generalize, resulting in overfitting where the model performs well on training data but poorly on unseen data. Additionally, in the context of anomaly detection, small datasets frequently suffer from class imbalance, with anomalies or rare events being significantly underrepresented compared to normal instances. This imbalance can skew the model's ability to detect anomalies accurately, as the classifier may become biased toward the majority class, leading to poor detection rates for the minority class. To address these challenges, techniques such as data augmentation [60] and transfer learning [61] are often employed. However, these strategies may not always be optimal and incorporating domain knowledge becomes crucial to enhance the effectiveness of supervised learning models with small datasets. Past research has highlighted the effectiveness of using frequency domain information to improve the separability and enhance the differentiation of datasets in welding technology [62]. Therefore, by leveraging domain-specific insights, it is possible to address some of the limitations associated with small datasets and improve model performance in anomaly detection tasks.

2.1.1 Logistic regression

Logistic regression is a widely used method for binary classification problems like anomaly detection [63]. Unlike linear regression, which predicts a continuous outcome, logistic regression is designed to predict the probability that a given input belongs to one of two classes (binary outcomes). The model is based on the logistic function, also known as the sigmoid function, to map any real-valued input number (x) into the probability (p) in range of 0 to 1, making it suitable for probability estimation using the Eq. 1. The weights (w) of the logistic regression can be obtained through gradient descent optimization [64, 65], aiming to minimize the cross-entropy loss.

$$p = \frac{1}{1 + e^{-wx}} \tag{1}$$

The decision function converts the probability value to 1 if $p \geq 0.5$ and to 0 otherwise; however, any other value can be used instead of 0.5. This algorithm allows to develop a linear regression based classified, and as for linear regression, it can be used for most of real-case scenarios, and the deployment and output computation are immediate.

2.1.2 Neural networks

If logistic regression does not yield satisfactory results, it may indicate that the classification problem is more complex and requires more advanced techniques. One of the most commonly used method, especially in welding, is the neural network. In a neural network [66], each neuron performs a nonlinear transformation of its inputs, and the network is structured in multiple layers. For binary classification problems, the output layer consists of a single neuron with a sigmoid activation function, similar to the logistic function. Gradient descent is employed in this case to minimize the cross-entropy loss function. Additionally, the multi-layered architecture typical of the neural network allows it to automatically learn weights across layers, providing enhanced feature selection and better separation of the feature space. However, unlike logistic regression, neural networks are often considered a "black box," making it difficult to interpret how the classification decision is made. Additionally, increasing model complexity by adding more neurons and hidden layers can lead to overfitting, where the model performs well on training data but poorly on test data, especially in cases with imbalanced datasets. Consequently, while neural networks offer generally more flexibility and better performance, they are more prone to overfitting and may underperform in comparison to logistic regression on certain datasets, especially for unbalanced ones typical of anomaly detection.

2.1.3 Decision trees and ensemble methods

A decision tree is a non-parametric, interpretable model that makes decisions by recursively splitting the data based on specific feature values. At each node in the tree, the algorithm selects the feature that best separates the data into classes based on a criterion like Gini impurity or information gain. The data is split into branches until a stopping condition is reached, such as all instances in a node belonging to the same class, or a maximum tree depth is achieved. One of the key advantages of decision trees is their interpretability. Each decision path from the root to a leaf node provides a clear, logical explanation for how the classification decision is made [67]. However, decision trees are prone to overfitting, especially when the tree becomes deep and complex. Aiming to improve the performance of a single decision tree an ensemble learning technique like random forest [68] can be used for the same scope. A random forest builds multiple decision trees on different subsets of the training data, generated through bootstrapping (random sampling with replacement). Each tree is built using a random subset of features, ensuring diversity among the trees. During prediction, each tree in the forest provides a classification, and the final output is determined by majority voting among the trees.

This ensemble approach reduces the risk of overfitting that individual decision trees face and generally improves the model's performance and robustness, particularly on complex datasets. Additionally, random forests are more resilient to noise and outliers, as the collective decision of multiple trees tends to average out the errors made by individual trees. Random forests offer a reliable and flexible approach for classification tasks when logistic regression or simpler models fall short, providing a balance between performance and interpretability.

2.2 Unsupervised learning methods for anomaly detection in small dataset

In contrast to supervised learning approaches, unsupervised learning techniques do not require large, labelled datasets. These methods can effectively handle unbalanced datasets or even single-class data to train algorithms. Unsupervised learning approaches can be classified into fully unsupervised and semi-supervised methods. The key difference lies in whether the algorithm is trained exclusively on normal behaviours data (semi-supervised) or on both normal and a small portion of anomalous data (unsupervised), where the majority of data represents normal behaviour.

Prominent algorithms used for unsupervised anomaly detection include reconstruction methods like PCA-based algorithms [69] and autoencoders (neural network-based) [70], density-based methods such as local outlier factor [71, 72], and tree-based methods like isolation forest [73]. These techniques can be applied in both unsupervised and semi-supervised contexts.

Although these methods are promising, especially in industrial applications, where they suit small and imbalanced datasets, they present several challenges. First, since unsupervised learning operates without labelled data, there is no clear ground truth to guide the learning process, making it difficult to evaluate model performance. Common supervised learning metrics, such as accuracy or precision, are not directly applicable, complicating the assessment and tuning of models. Additionally, unsupervised algorithms often struggle with noisy data because they lack labels to differentiate noise from meaningful patterns.

As a result, unsupervised methods typically yield poorer performance compared to supervised learning, particularly in detecting slight variations from normal parameters, which can be critical in welding processes. These algorithms are designed to identify rare events rather than subtle deviations that may still indicate potential defects. Feature extraction becomes crucial in this context, as it can help capture local signal behaviours, emphasizing differences and improving model performance [62].

2.2.1 Reconstruction-based anomaly detection: PCA and autoencoder

Anomaly detection using dimensionality reduction techniques often begins with a coding-decoding process that relies on training models exclusively with normal data. The idea is that by learning to compress and then reconstruct normal data, the model captures essential patterns of normality in the dataset. When new data deviates from these learned patterns, the reconstruction error increases, indicating a potential anomaly. By setting a threshold on this reconstruction error, anomalies can be detected whenever the error exceeds this predefined level.

Principal component analysis (PCA) and autoencoders are two common methods for anomaly detection. PCA reduces data to a lower-dimensional linear space, identifying anomalies via significant reconstruction errors, which indicate deviations from typical variance patterns. Autoencoders, using neural networks, perform non-linear dimensionality reduction by compressing data into a latent space and reconstructing it. Their ability to capture complex patterns makes them effective for intricate datasets.

2.2.2 Density-based anomaly detection: local outlier factor

Local outlier factor (LOF) method is a density-based anomaly detection technique used for identifying unusual data points by comparing the density of data around each point to the density around its neighbours. The basic idea is that normal points usually have a similar "density" of nearby points, meaning they are in areas with other points close by. Anomalies, on the other hand, often have fewer or more distant neighbours, making them stand out in terms of density, especially if the pattern of normal density is learnt using only normal data. LOF calculates a score based on how isolated a point is from its neighbours, which, in an online process monitoring context, requires the agent to store the training dataset in memory. When a point has a much lower density compared to its neighbours, it receives a high LOF score, identifying it as an outlier. While this approach is effective for datasets with multiple regions of varying densities and can handle complex data structures, it is computationally intensive. This is because it requires storing all normal data in memory and continuously computing distance metrics.

2.2.3 Isolation forest

The isolation forest algorithm is an ensemble-based, unsupervised anomaly detection method that identifies anomalies by recursively partitioning the data using multiple trees, similar to random forest. The key idea is that anomalies are easier to isolate than normal data points. The algorithm builds binary trees where each node randomly selects a feature and splits it at a random value, creating subsets of the data. This process continues until each point is isolated in a leaf node, or a maximum tree depth is reached. Once the splits are learned from the normal data, the algorithm can score new, online samples from the process to identify anomalies.

Unlike LOF, which requires preprocessing like standardization, the isolation forest's tree-based structure naturally accommodates variations in feature scales, making it efficient for high-dimensional datasets without the need for normalization, as happen for the others tree-based algorithms.

2.3 Experimental setup and data collection

The experimental setup for this study, illustrated in Fig. 3 is composed by an Omron Viper s850 robotic arm, a Miller Phoenix 456 power supply and a multi-sensor acquisition system. The welding current and the welding voltage were collected at a sample rate of 5 kHz using a NI 6361 acquisition board and employing different process parameters for each wall as outlined in Table 2. During the experimental campaign, 15 walls, each consisting of 10 layers with a length of 60 mm, were deposited to study the stability of WAAM for Invar 36. The goal was to assess the optimal parameters, which correspond to the parameters used for the wall no. 4. The combination of process parameters assured the dip transfer process for the droplet to the melting pool. The gas flow rate, which consist of pure argon, and the contact tip to workpiece distance (CTWD) were fixed to 18 L/min and 13 mm, respectively. The CTWD was chosen to ensure adequate shielding of the melting pool, preventing defects such as porosity or explosive behaviour. The experimental design was based on a Latin hypercube sampling approach, considering different levels of welding speed, wire feed speed, and welding voltage. Specifically, six levels were set for wire feed speed, four for welding voltage, and four for welding speed. From this initial experimental plan, extreme conditions—such as high feed rate with low welding voltage and similar—were excluded, allowing us to select the most suitable combinations for our study based on authors previous experience.

Table 2 Design of experiments of natural dip transfer Invar 36 deposition

Wall no	Wire Feed Speed [m/min]	Welding voltage [V]	Welding Speed [mm/s]
1	3.5	19	3.2
2	4	19	4.5
3	3.5	22.5	2.0
4	3.5	21	4.0
5	3.5	19	4.5
6	3.5	20	2.0
7	3	19	3.2
8	2.5	21	3.2
9	2	20	4.5
10	6	19	4.5
11	6	21	3.2
12	4	22.5	4.5
13	3	22.5	4.0
14	4	19	4.0
15	3	20	4.5

Aiming to label the collected samples, a welding camera (Xiris XVC-1000) was employed to capture images of the process, and the labels were manually assigned using the

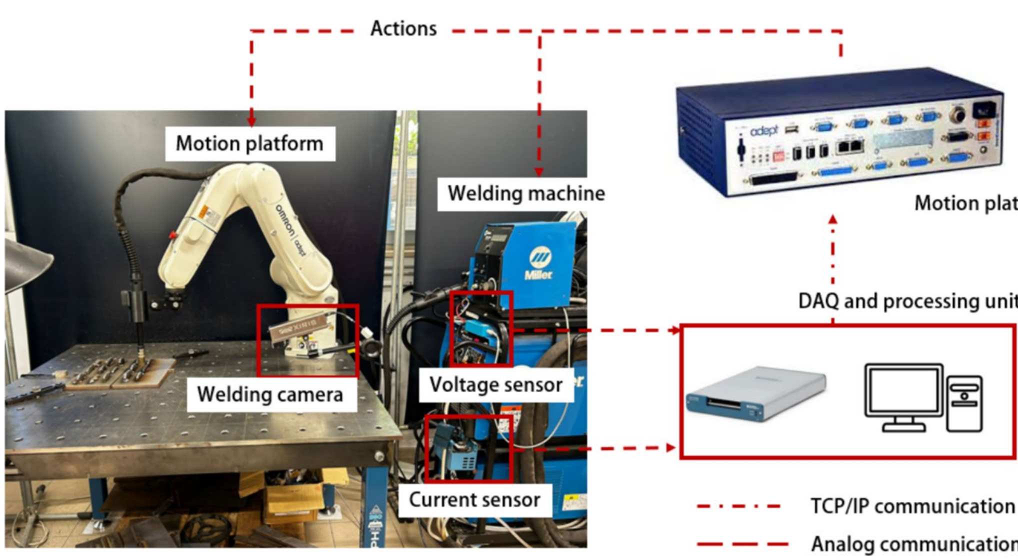


Fig. 3 Experimental setup employed in this work

collected images, the welding sound, and the surface appearance of the deposited layer. Aiming to develop an online software module, the collected samples were divided in windows of 1 s length, which is a usual procedure for welding process monitoring [74]. Figure 4 shows a typical 1 s length window associated to a good deposition data, which shows the dip transfer process in both welding current and welding voltage signals. Following data collection and labelling, an unbalanced dataset was obtained, consisting of 1698 samples, of which only 20%—a total of 344 samples—represent various anomalies. These anomalies include porosity, layer collapse, arc start issues, excessive spatter, and excessive humping and some examples of the obtained walls with some defects introduced by process anomaly are shown in Fig. 5. Figure 6 displays examples of collected signals associated with normal deposition or "in-control" process and signals associated with anomalous deposition, specifically instability, which is considered an out-of-control condition.

During the experimental tests, various anomalies were identified. Despite their differences, these anomalies were grouped under the same labels to simplify the anomaly detection process. This was essential, as the number of distinct classes was insufficient for supervised learning without such grouping. The identified anomalies included defects such as pool shift, layer collapse, and issues related to incorrect heating and material properties. In particular, pool shift defects occur due to the distortion of the electric field around the welding arc, leading to problems like layer collapse, where there is inadequate material support for the printing process, or deviations from the desired weld pattern. Excessive humps, often caused by improper heat input, can result in undercut. Additional anomalies were associated with layer collapse caused by excessive substrate heating, which was exacerbated by the thermal properties of Invar 36 alloy. Moreover, porosity and explosive arc behaviour were observed, typically resulting from incorrect parameter selection or insufficient shielding, particularly due to an improper CTWD.

2.4 Analysis of welding signals and feature extraction

In this study, we compare the results obtained from both supervised and unsupervised learning approaches, using the

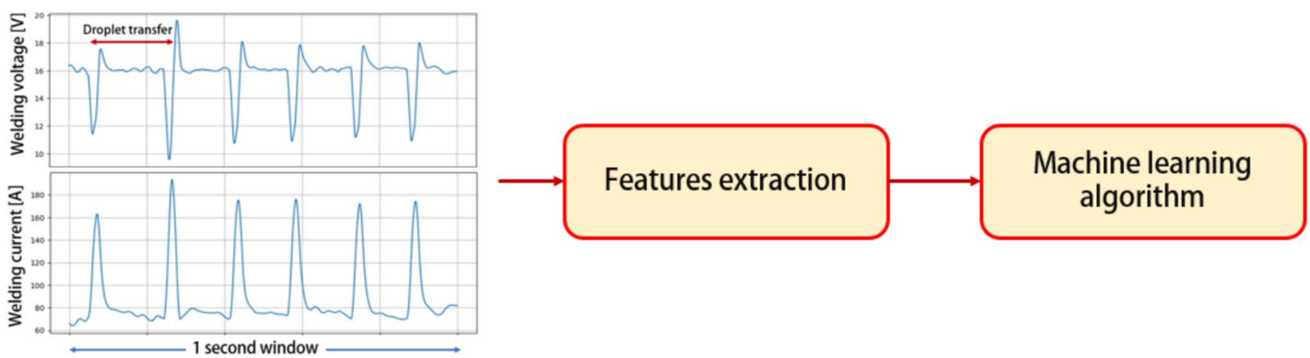
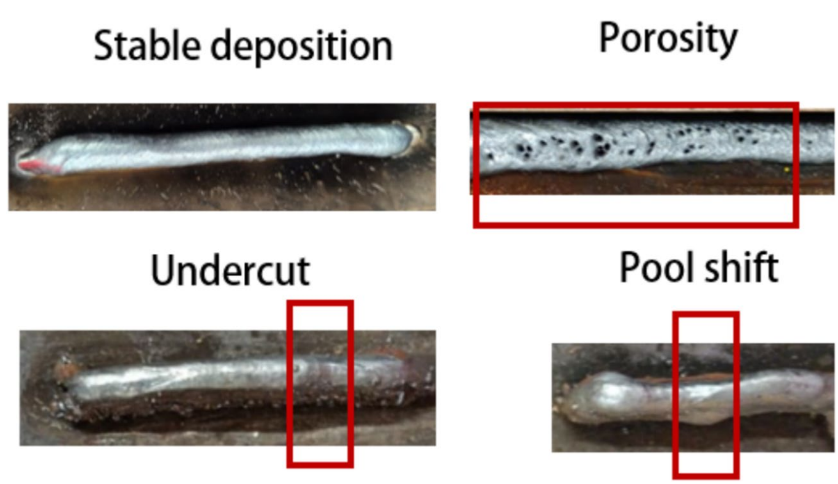


Fig. 4 The samples used in this study to develop the anomaly detection module consist of 1 s length welding voltage and welding current signals, from which features have been extracted to be used in input to ML algorithms

Fig. 5 Images of deposited walls from the experimental campaign, showcasing both good-quality and defective deposition data, influenced by anomalies in welding current and voltage signals



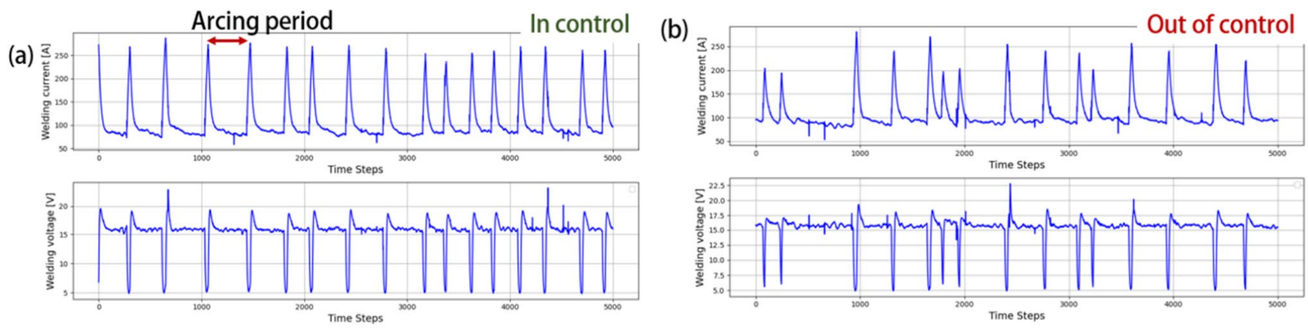


Fig. 6 **a** Welding current and voltage signals related to in-control (or good) deposition, compared to **b** signals associated with out-of-control deposition, such as spattering during the deposition process

same set of input features. These features were extracted from the time and frequency domains of both welding current and welding voltage signals, as recommended by the literature. For the time-domain signals, features such as energy, standard deviation, median, skewness, and kurtosis were calculated over 1-s windows, resulting in 10 features. Additionally, features such as average pulse time, the number of peaks, and the maximum values were extracted from the welding current, as shown in Fig. 7a, for a total of 13 features. Then, the Morlet transform—a tool for visualization and analysis of the signals in the time–frequency domain—is applied and shown in Fig. 7b to analyze the frequency response of the signals, revealing the frequency bandwidth relevant to the process parameter, between 0 and 1500 Hz.

Figure 8 shows two scalograms, one corresponding to a good deposition and the other, due to a process anomaly, showing defects such as porosity. The Morlet transform allowed us to study in the time–frequency domain the characteristics of the signals, highlighting the importance of information in the lower frequency bands for assessing deposition quality in natural dip transfer processes, complementing time-domain. Figure 9 shows the time domain values of current and voltage associated to several anomalies collected during the experiments. Figure 10 presents the FFT responses from both anomalous processes—such as poor gas, porosity, and excessive spatter—as well as from normal deposition data. This is obtained removing the mean from the time series window signals aiming to reduce the impact of the DC component on the FFT response. The results show how varying process parameters affect the frequency response, which remains within the 0–300 Hz range due to the dip transfer process used. Additionally, distinct

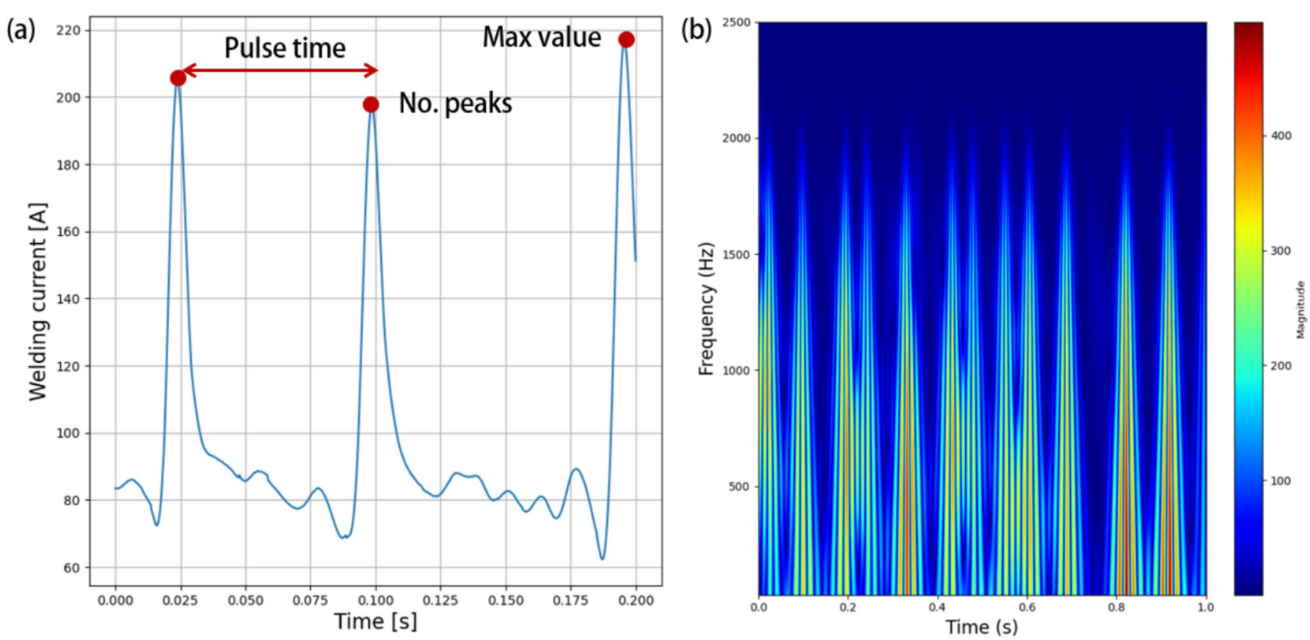


Fig. 7 **a** Features extracted from the time domain window and **b** the time–frequency response computed by Morlet transform

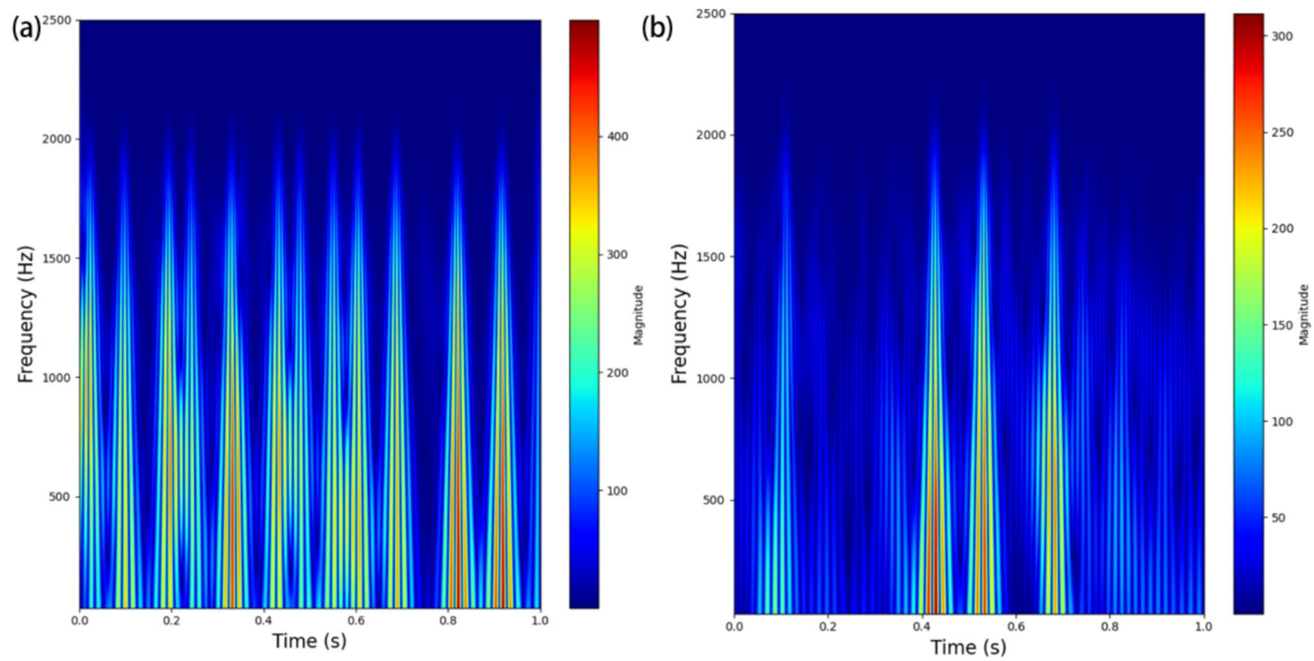


Fig. 8 Morlet transform for **a** normal deposition signal and **b** anomalous deposition signals which highlight the differences between signals also in the frequency domain

phenomena are highlighted, revealing complex characteristics in the frequency domain that help differentiate the various process conditions.

From the FFT response of both welding current and welding voltage signals, the extracted features include energy, standard deviation, median, skewness, and kurtosis, for a total of 23 features.

To extract meaningful features from the frequency domain, we performed a 5-level wavelet decomposition using a 4th-order Daubechies wavelet. This allowed us to capture features across the following frequency bands: 2500–1250 Hz, 1250–625 Hz, 625–312.5 Hz, 312.5–156.2 Hz, 156.2–78.1 Hz, and 0–78.1 Hz. Previous study highlighted how this wavelet shape and order can be used to better separate the datapoints in the features space for different materials and welding processes [62]. Figures 11 and 12 show the decomposition levels obtained for different samples in the dataset. Additional analysis conducted during this work indicates that wavelets such as Haar are not suitable for this purpose, as they do not provide distinct information across different decomposition levels. Aiming to maintain as much possible information, energy, standard deviation, median, skewness, and kurtosis, maximum and minimum values are extracted from both welding current and welding voltage decomposition levels, for a total of additional 60 features, for a total of 83 features collected in both time and frequency domain. The extracted features are reported in Table 3.

As highlighted in the detailed analysis of the collected signals, both the global frequency response, obtained via FFT, and the local frequency response within specific bandwidths, captured through DWT, provide significant insights in the time–frequency domain. Additionally, information from the time domain was also considered. From this analysis, we extracted 83 features that enable a comprehensive evaluation of 1 s of deposition, encompassing all key characteristics in the frequency, time–frequency, and time domains. We observed that signals associated with different anomalies exhibit distinct behaviours, not only in the time domain. Therefore, feature engineering is crucial for the effective development of machine learning algorithms, particularly for unsupervised learning.

After feature extraction, the dataset containing only data from successful depositions was divided into training and validation sets in a 75–25 ratio, resulting in 1083 samples for training and 271 for validating normal behaviour. An additional 384 samples associated with anomalous conditions were combined with the validation set to evaluate the performance of different online anomaly detection methods. Figure 13 shows PCA results, highlighting that simple linear dimensionality reduction allows for easy detection of some outliers. However, for other samples, the feature characteristics closely resemble those of the training data—particularly for instabilities like humping, weld pool shifts, and spatter. As a result, more advanced techniques are required to detect these complex anomalies.

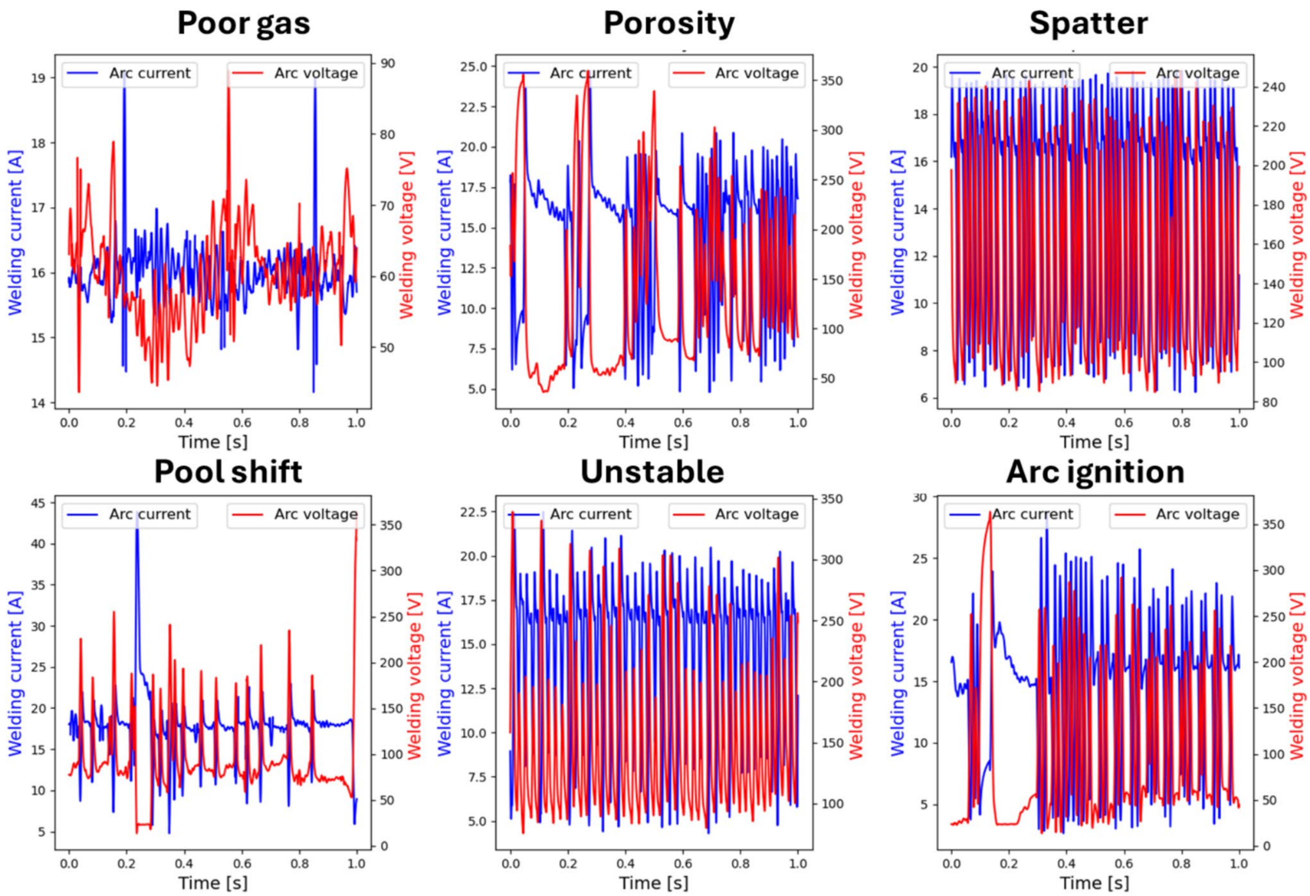


Fig. 9 Time domain signals related to process anomalies collected during the experimental campaign for Invar 36

2.5 Contextualizing process monitoring and quality assessment under variable certification standards

If a waveform appears unusual or "not fine" under a certain set of process parameters, it does not necessarily indicate a defect or a true anomaly in the component. This is because the same waveform shape could be considered normal when seen under a different set of parameters. In other words, a waveform that seems anomalous within one parameter range might represent acceptable behaviour in another range.

Consequently, there are situations where the process deviates from strictly qualified parameters—those that are typically required to produce high-quality components that meet specific standards. However, components produced under such conditions may still be of acceptable quality, even if they fall outside the optimal parameter range specified by quality standards. This complexity highlights the need to consider the context of process parameters when assessing waveform anomalies, as not all deviations from ideal conditions lead to defective parts.

At this stage, it is important to contextualize the application of process monitoring based on the required quality standards. When strict certification standards are in place for product production, it is essential to assess product quality specifically against those certified parameters. To address this, previous work has proposed methodologies that monitor only in relation to certified parameters, making this approach more applicable from an industrial perspective.

In scenarios where certification standards are stringent, supervised learning techniques are not suitable, as it is not feasible to apply a single-class model in a supervised framework. In such cases, the appropriate approach is to use unsupervised learning methods, which do not require predefined labels and are well-suited for identifying deviations in a strict quality-controlled environment.

On the other hand, for applications where strict certification is not mandatory or where the aim is to evaluate general process quality, utilizing the full dataset collected during process certification procedure can be valuable. This is because a process may contain anomalies relative to optimal cases, yet these anomalies do not necessarily indicate defects or low-quality characteristics. Parts may

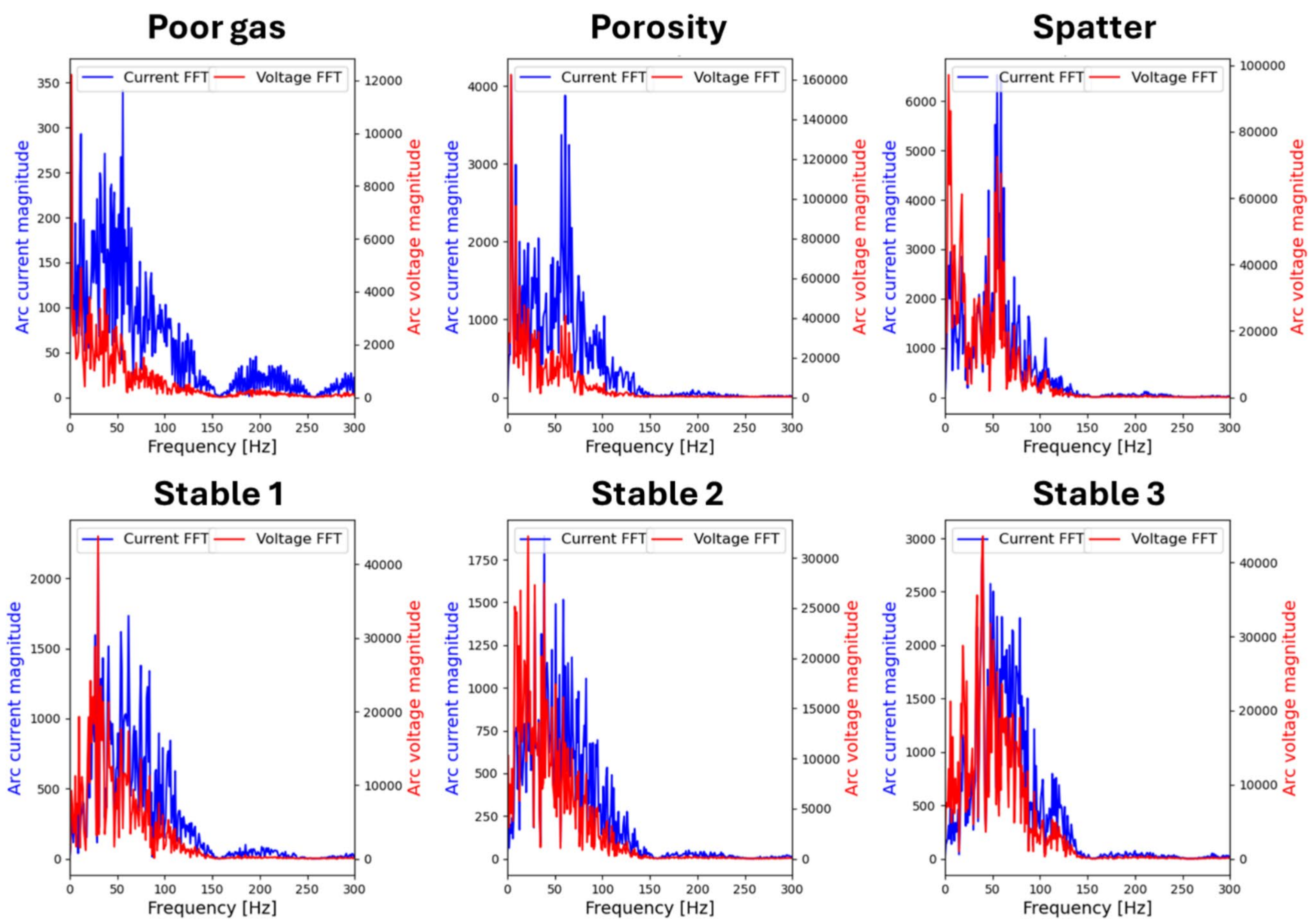


Fig. 10 FFT response of both welding current and welding voltage signals related to stable deposition conditions and process anomalies collected during the experimental campaign

still be usable in applications with more flexible certification requirements. Thus, when certification criteria are less rigid, using all available data to evaluate general component quality can provide broader insights, particularly when informed by experience collected during the certification process.

2.6 Metrics to evaluate performance in small and unbalanced datasets

When data are collected from a qualification procedure, it is likely that the dataset will be imbalanced, as anomalies are often rare. Consequently, using the accuracy as a performance metric can be misleading. Instead, metrics like precision (p), recall (re), and the F-score provide a more precise evaluation when the goal is to solve binary classification problems like anomaly detection. Precision measures the proportion of correctly identified anomalies out of all predicted anomalies, while recall captures the proportion of actual anomalies that were correctly detected. The F-score, as outlined in Eqs. 2–4, balances these two metrics. In these equations, true positives (TP) refer to correctly identified anomalies, true negatives (TN) to correctly identified normal cases, while false positives (FP) represent false alarms, and false negatives (FN) represent missed anomalies. Finally, when missing an anomaly is more critical than having a higher false alarm rate—such as in AM, where any anomaly can potentially result in a defect—the F2 score, reported in Eq. 5, becomes an important metric to consider, which places more emphasis on recall compared to precision. The metrics employed in this study are summarized in Table 4.

$$p = \frac{TP}{TP + FP} \tag{2}$$

$$re = \frac{TP}{TP + FN} \tag{3}$$

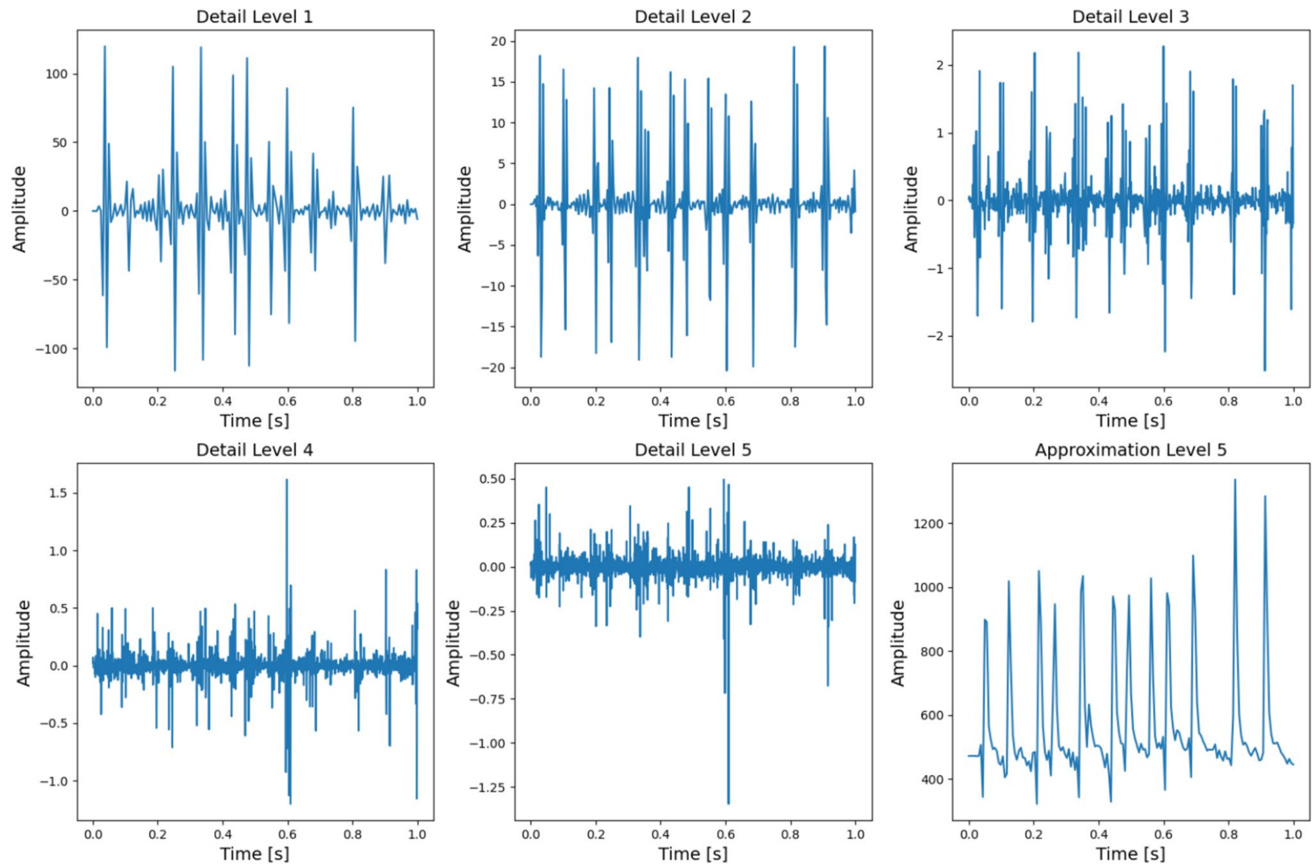


Fig. 11 Decomposition levels obtained through a 5-level Daubechies wavelet decomposition for a signal associated to a stable deposition process previously plotted as "Stable 1"

$$F1score = \frac{(1+\beta^2)\cdot p \cdot re}{\beta^2 \cdot p + re} = \frac{2p \cdot re}{p + re} \tag{4}$$

$$F2score = \frac{(1+\beta^2)\cdot p \cdot re}{\beta^2 \cdot p + re} = \frac{5p \cdot re}{4p + re} \tag{5}$$

As a result of contextualizing process monitoring, the most appropriate metrics can be selected. While process monitoring is valuable in industrial anomaly detection, especially in situations where labelled data is scarce or expensive to obtain, these approaches are inherently prone to false alarms (Type I errors) and missed anomalies (Type II errors) [75], both of which can degrade system performance. High false alarm rates may lead to unnecessary production stoppages, increased downtime, inspection costs, and maintenance interventions. These disruptions not only raise operational expenses but can also cause inefficiencies in supply chain management. If the total cost of addressing false alarms surpasses the cost of producing and discarding defective parts, the detection system becomes economically unfeasible. Therefore, it is crucial to contextualize the metrics in relation to the production environment (e.g. the material employed) and the stringent certification processes in place. For example, given the high cost of Invar 36 wire (approximately 100 USD/kg), it is essential to prioritize detecting anomalies to avoid missing them. An error occurring after just 1 h of deposition could lead to a waste of around 1000 USD.

3 Results and discussion of supervised learning-based process anomaly detection

In this study, four distinct supervised machine learning models were utilized to detect anomalies in the WAAM process, using features extracted from both the time and frequency domains of welding current and voltage signals. The models employed include logistic regression, random forest, decision tree, and neural network classifiers. Each model was trained with optimized hyperparameters, determined through a trial-and-error approach, and the best-performing results are presented.

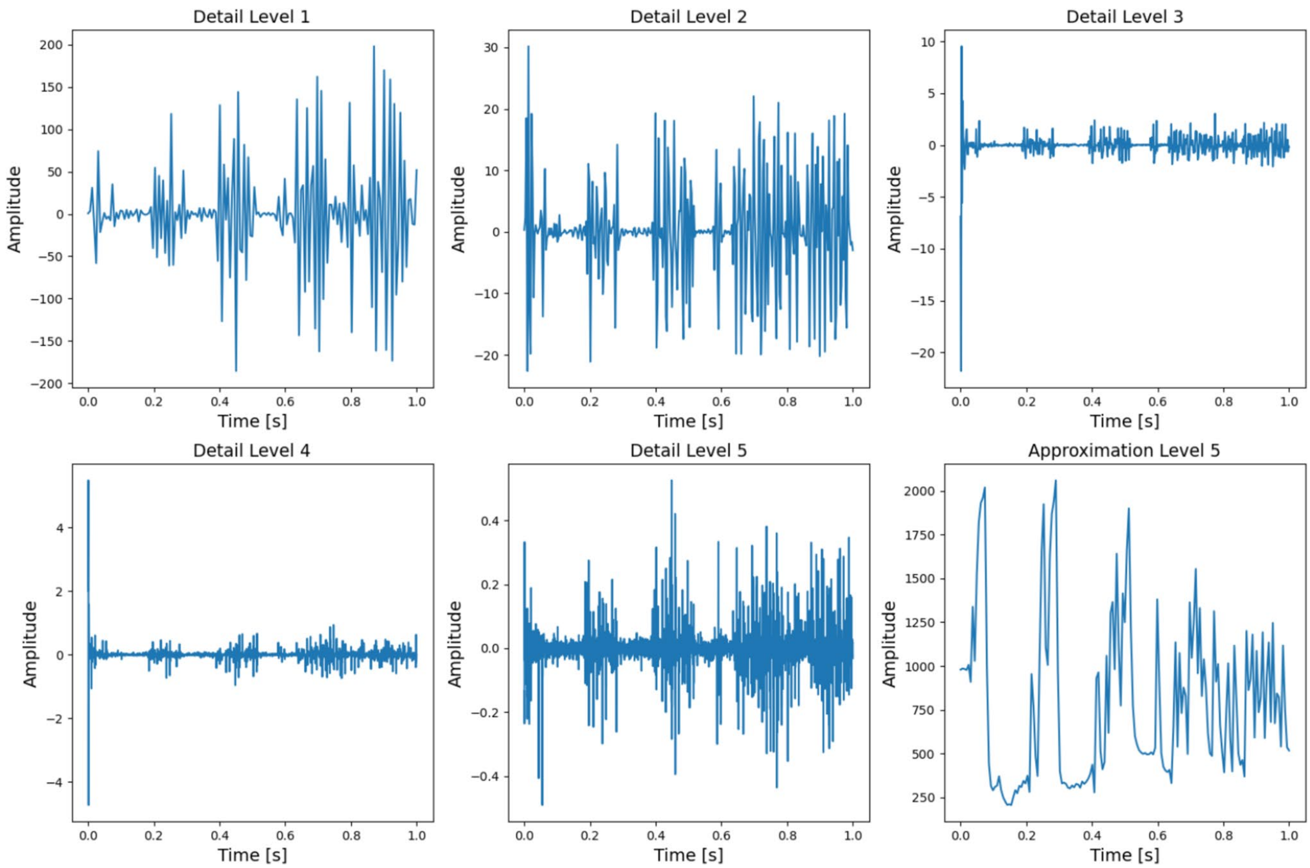


Fig. 12 Decomposition levels obtained through a 5-level Daubechies wavelet decomposition for a signal associated with porosity

Table 3 Extracted features from welding current and welding voltage signals

Time domain welding current	Time domain welding voltage	FFT features in the range 0–300 Hz for both welding voltage and current	DWT features for both welding voltage and current
Energy	Energy	Energy	Energy
Variance	Variance	Variance	Variance
Skewness	Skewness	Skewness	Skewness
Kurtosis	Kurtosis	Kurtosis	Kurtosis
Median	Median	Median	Median
Number of peaks	-	-	-
Maximum value	-	-	-
Average pulse time	-	-	-

3.1 Hyperparameters of the employed models

All the models have been implemented using the python library Scikit-Learn [76]. The logistic regression model was selected for its simplicity and interpretability, making it an ideal baseline model for binary classification tasks. It was configured with a maximum iteration limit of 1000 to ensure convergence during training, and the *liblinear* solver was chosen due to its suitability for small datasets and binary classification problems. The regularization parameter C was set to 1.0 to balance the trade-off between underfitting and overfitting, while the random state was fixed to ensure reproducibility of the results. The random forest model was chosen for its ability to handle high-dimensional data. The model was configured with 50 trees, which strikes a balance between computational efficiency and model performance.

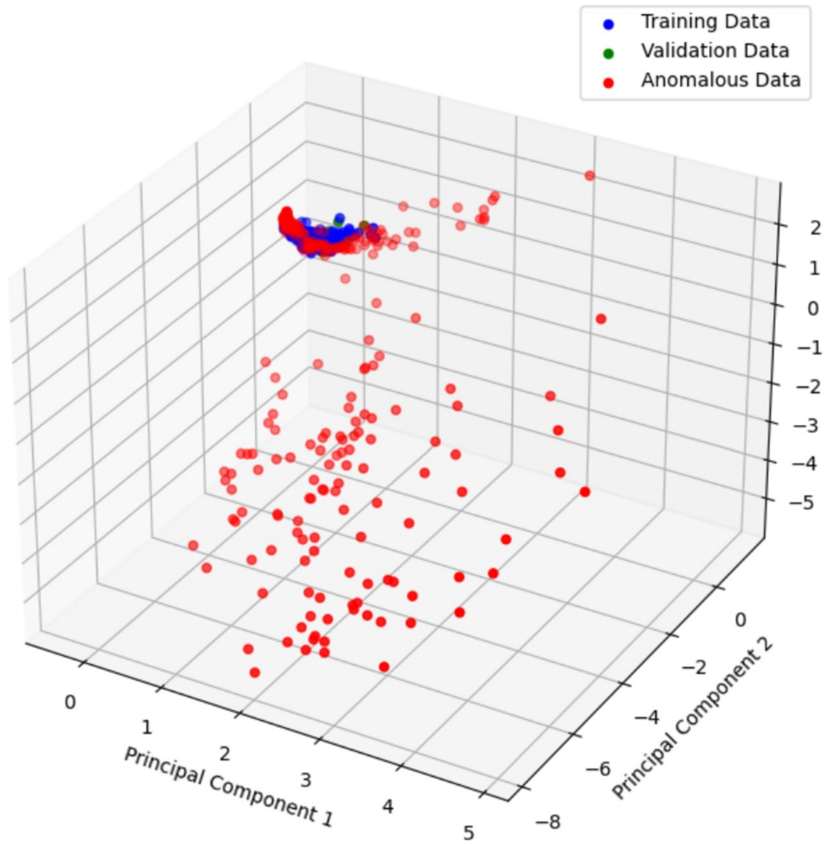


Fig. 13 The dimensionality reduction utilized for visualization reveals that certain anomalies, such as humping, pool shift, and spatter, are challenging to detect, as their signal signatures closely resemble those observed under normal conditions. Unlike rare events like porosity, arc ignition failures, or insufficient shielding gas, these anomalies do not stand out as easily

Table 4 Summary of performance metrics

Metrics	Meaning
Precision	The proportion of true anomalies out of all predicted anomalies. Low precision indicate high false alarm
Recall	The proportion of true anomalies out of all actual positive instances. Low recall indicate low performance in detecting anomalies (missed anomalies)
F- score	The harmonic mean of precision and recall. With $\beta = 2$ more importance ed is given to recall (improve false alarm better then miss anomalies)
Accuracy	The proportion of correctly classified instances
True positive	Instances correctly predicted as anomalies
True negative	Instances correctly predicted as normal behaviour
False negative	Instances incorrectly predicted as normal behaviour (missed anomaly)
False positive	Instances incorrectly predicted as anomaly (false alarm)

To reduce the risk of overfitting, the trees were restricted to a maximum depth of 4. Additionally, a minimum of 4 samples was required to split a node, and at least 2 samples were needed for a leaf node. The number of features considered for each tree split was limited to 9, helping to reduce model variance. For the decision tree model, a similar approach to random forest was employed, but with a single tree rather than an ensemble. This configuration allows for a more interpretable model, though it is more prone to overfitting compared to random forest. The neural network model featured a single hidden layer with 8 neurons, providing a reasonable starting point for this binary classification problem. The ReLU activation function was used for the hidden layer due to its efficiency and ability to mitigate the vanishing gradient problem. The Adam optimizer was selected with a constant learning rate of 0.00001 and a maximum of 60,000 iterations.

Both logistic regression and neural network models utilized standardized input features, as this scaling approach is known to improve their performance by ensuring the models are not biased by varying feature magnitudes [77].

3.2 Dataset generation for supervised learning

After collecting all the data for each second of deposition and extracting the features, the initial dataset consists of 1698 samples with 83 features each. For a supervised learning approach, the dataset was split into training and testing sets using an 85–15% ratio. This resulted in 1443 training samples, comprising 1151 defect-free cases (80%) and 292 anomalous cases (20%). The test set contains 275 samples, with 203 defect-free cases (74%) and 72 anomalous cases (26%). After splitting the dataset, no further preprocessing was applied to the features for tree-based algorithms, as these models can handle raw feature scales effectively. However, for logistic regression and neural networks, the features were normalized to a range of 0–1 using Eq. 6.

$$n_f = \frac{old_f - \max(f)}{\max(f) - \min(f)} \tag{6}$$

3.3 Discussion of the results for supervised learning

The logistic regression and neural network models achieved an accuracy of 95.3% and 96.8%, while random forest achieves a higher accuracy compared to decision tree with 95.7% against 93.7%. However, due to the unbalanced nature of the dataset, evaluating the performance of the models requires additional metrics. Regarding the logistic regression, the model shows the highest precision of 100% with a recall of 76.9% reflecting its effectiveness in identifying anomalies without false alarms. Figure 13 shows the confusion matrix for all the proposed models, which shows for logistic regression that 12 samples were incorrectly classified as normal.

Regarding the other models, decision tree presents the lower F1-score of 82.9% introduced by the lower recall of 75% and the reduced ability to recognize normal cases with respect to other models due to the precision of 92.8%. For what concern the random forest model, it reduces the number of missing anomalies, with an improvement in the recall with respect to decision tree to 86.5% but with a slightly higher number of false positive, with a precision of 91.8% with a general F1-score of 89.1%. Finally, the neural network model represents the best solution with an F1-score of 92.1%, thanks to reduced missing anomalies and reduced false positive with a precision and a recall respectively of 94% and 90.4%, which represent the best-balanced case (Fig. 14).

Despite this, it is important to consider also the high interpretability of both random forest and logistic regression which may allow for a root cause identification and further analysis of the obtained results, which is more complex to reach via black-box models like neural networks.

Finally, as discussed in the previous section, supervised machine learning models are prone to overfitting. To empirically illustrate this phenomenon, Fig. 15 presents the learning curve of the reported neural network. The figure demonstrates that from approximately epoch 2000 onwards, a noticeable divergence occurs between the training and test loss, indicating that the model is memorizing training data rather than generalizing effectively. The use of an early stopping technique can improve results by preventing overfitting. However, the results in Table 5 show a reduction in performance due to this technique, particularly with regard to precision, indicating that more false alarms were detected. The results are summarized in terms of precision, recall, F1-score, F2-score, and accuracy in Table 5.

4 Results and discussion of unsupervised learning-based process anomaly detection

In the final part of this study, three distinct unsupervised machine learning models were employed to detect anomalies in the WAAM process. These models utilized features extracted from both the time and frequency domains of welding current and voltage signals. The approaches included a density-based method using the local outlier factor, a tree-based method with isolation forest, and a reconstruction error-based method leveraging PCA. Each model was trained using hyperparameters optimized through a trial-and-error process, and also, in this case, the best-performing results are reported.

4.1 Hyperparameters of the employed models

The local outlier factor (LOF) model was configured with 30 neighbours to estimate local density, enabled novelty detection to handle unseen data, and assumed a contamination level of 1%. This contamination parameter indicates the expected proportion of anomalies in the dataset and is critical for defining the decision threshold for outlier classification. In models like isolation forest and LOF, the contamination level balances sensitivity to anomalies against the risk of false positives. Choosing an appropriate value ensures the model aligns with the true anomaly distribution: underestimating it may miss anomalies, while overestimating it could lead to excessive false alarms. Although the dataset consists solely of defect-free data, selecting an optimal contamination level is still important to fine-tune the results. In the proposed approach, this value is set within the range of 1–5%.

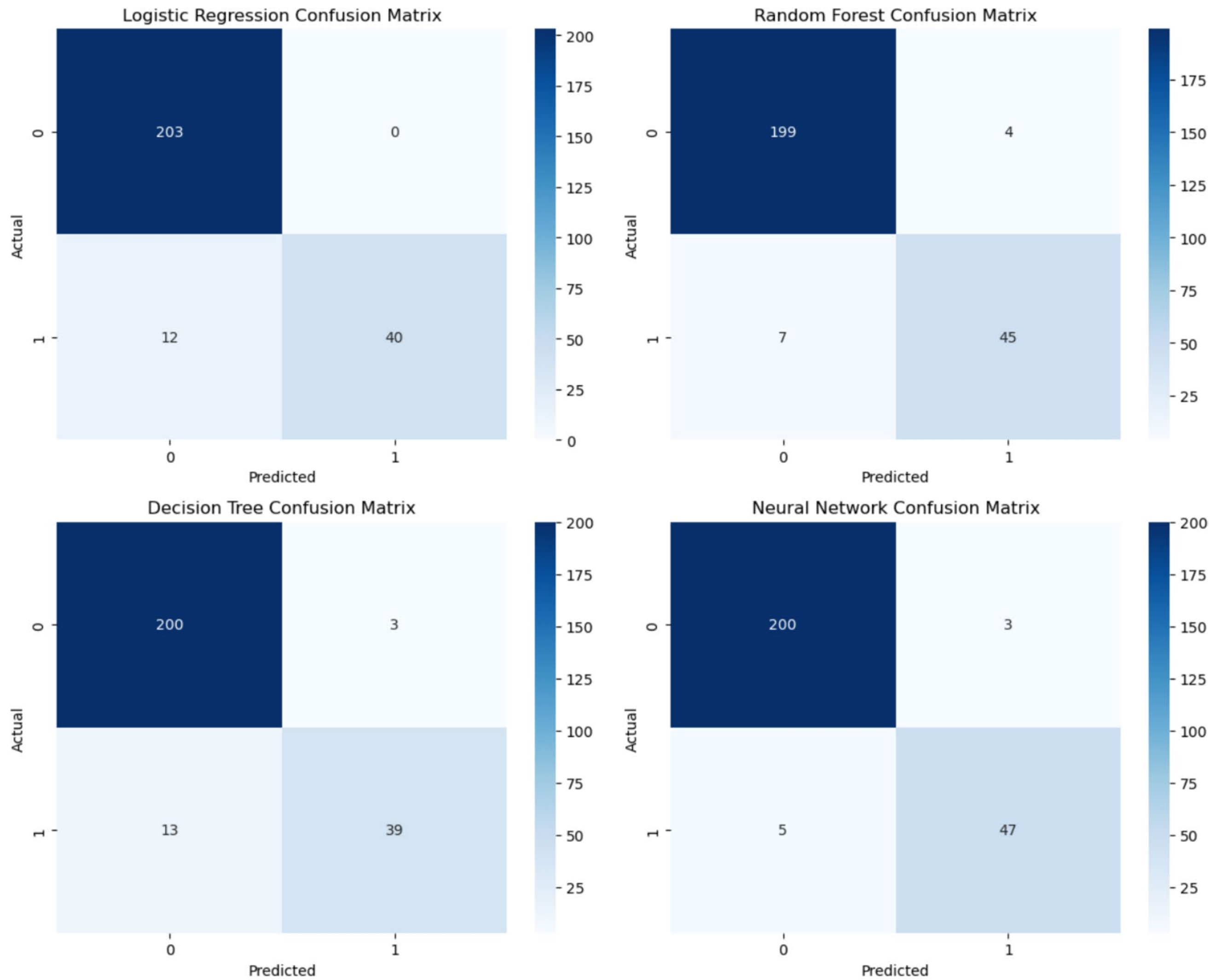


Fig. 14 Confusion matrices as results on test dataset

If the dataset contains more anomalies, and their exact proportion is known—as in a supervised approach—this value can directly inform the contamination parameter. The isolation forest model, on the other hand, was configured with 50 trees and a contamination level of 5%. For the principal component analysis (PCA) model, three components were used to retain sufficient variance for effective reconstruction. A contamination level of 5% was also employed, setting the anomaly detection threshold at the 95th percentile of the training reconstruction errors.

4.2 Dataset generation for unsupervised learning

After collecting the data and extracting features, the dataset was divided into anomaly-free deposition data and anomalous data. Using an 80–20% split, the anomaly-free data were further split into training data and testing data, which were combined with the anomalous data for model evaluation. Specifically, 1083 samples were used for training, while the remaining 271 anomaly-free samples and 344 anomalous samples were combined, resulting in a total of 615 samples for testing. No additional preprocessing was applied to the features for the isolation forest and PCA algorithms. However, for the local outlier factor algorithm, the features were normalized to a range of 0 to 1.

4.3 Discussion of the results for unsupervised learning

As discussed, two distinct approaches can be employed to develop unsupervised learning applications. The first approach involves using only defect-free deposition data to detect anomalies. Following this criterion, Fig. 16 presents

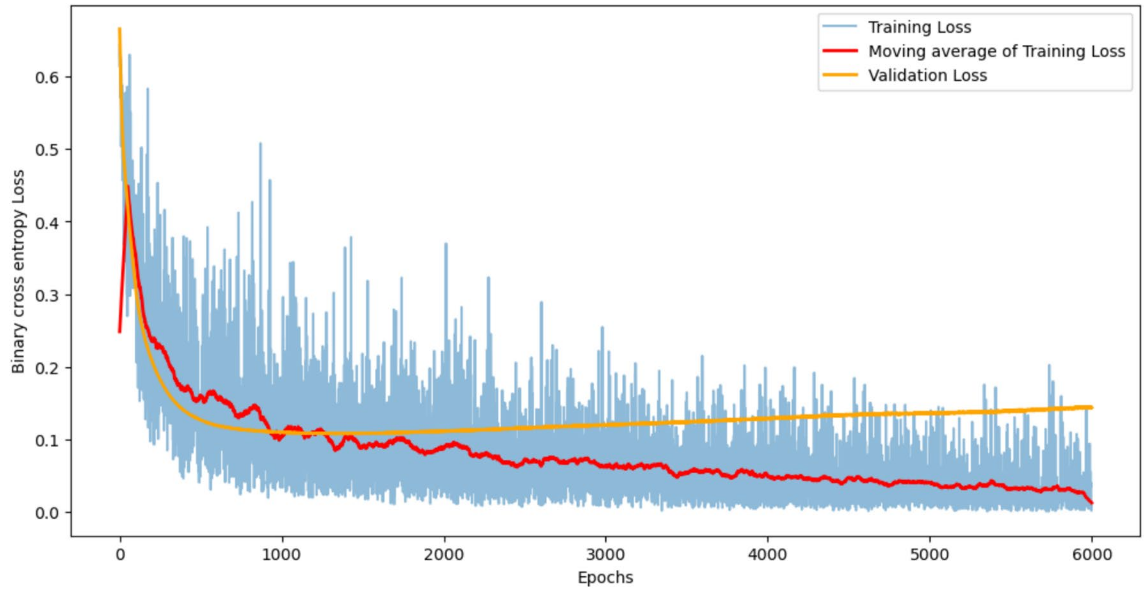


Fig. 15 Learning curves of neural network which demonstrate the overfitting which affect these algorithms

Table 5 Comparison of the results obtained using supervised machine learning algorithms

Model	Precision	Recall	F1-score	F2-score	Accuracy
Logistic regression	100.0	76.9	86.9	80.6	95.3
Random forest	91.8	86.5	89.1	87.5	95.7
Decision tree	92.8	75.0	82.9	78.0	93.7
Neural network	94.0	90.4	92.1	91.1	96.8
NN with early stopping	86.53	86.53	86.53	86.53	94.5

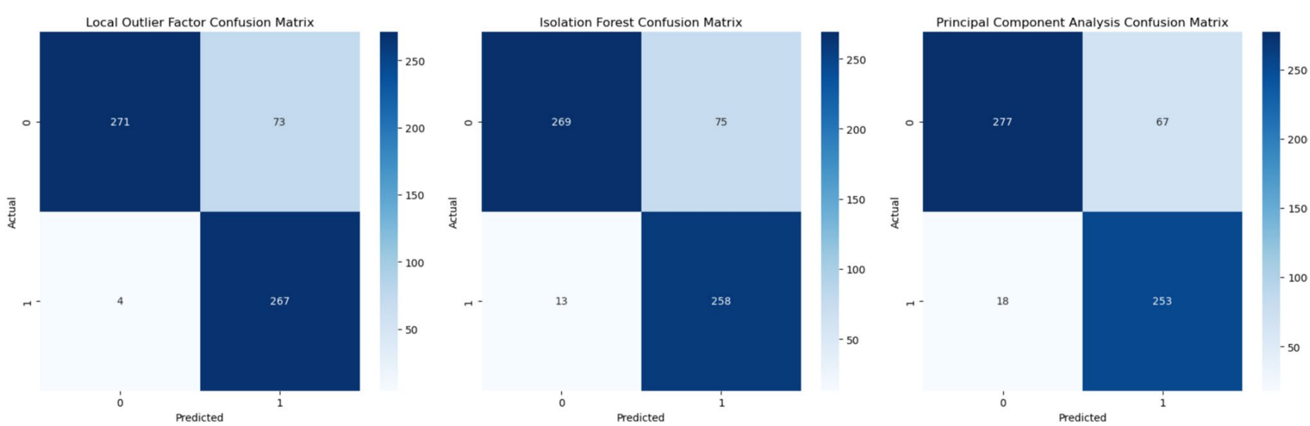


Fig. 16 Confusion matrices as results on test dataset composed by both anomalous and good deposition data, once the models have been trained using only good deposition data

the confusion matrices for various models evaluated on the test dataset, as described earlier.

The results are summarized in terms of precision, recall, F1-score, F2-score, and accuracy in Table 6. Compared to the results observed with supervised learning, the overall accuracy of the algorithms decreases due to reduced generalization ability of unsupervised learning models. The highest accuracy, 87.5%, is achieved by the LOF algorithm, while isolation forest and PCA reach 85.7% and 86.2%, respectively. However, the recall values for unsupervised learning are notably higher than those reached by supervised learning algorithms. LOF achieves a recall of 98.5%, followed by isolation forest at 95.2% and PCA at 93.3%. These values surpass the maximum recall observed with neural networks in supervised learning, which was 90.4%. This indicates that using unsupervised learning with only defect-free data allows the system to detect real anomalies more reliably during online monitoring. However, the reduced precision values suggest a higher false alarm rate compared to supervised learning. LOF achieves a precision of 78.3%, isolation forest 77.5%, and PCA 79.1%.

As a result, the F2-scores in this approach are higher than the F1-scores, contrasting with the trends observed in supervised learning. The F1-scores are 87.4% for LOF, 85.4% for isolation forest, and 85.6% for PCA. Meanwhile, the F2-scores are 94.7% for LOF, 91.0% for isolation forest, and 90.0% for PCA.

An alternative approach, distinct from the one-class learning method used in our previous research, involves utilizing both anomaly-free and anomalous deposition data. In this case, using the same dataset as in supervised learning, 20% of the data in the training set was labelled as anomalous deposition data, resulting in a contamination level of 20% for the unsupervised learning models previously discussed. By applying the same hyperparameters as before, the results shown in Table 7 and in Fig. 17 were obtained.

Compared to the previously discussed results, precision is significantly improved, leading to a higher F1-score. Notably, the isolation forest model shows a substantial increase, rising from 85.4 to 92.0%, which represents the best F1-score among all the models tested, matching the results achieved by the neural network. However, the reduced recall relative to other unsupervised methods leads to a slight decrease in the F2-score, which still matches the neural network's best result of 91%. Therefore, the proposed methodology demonstrates that isolation forest achieves balanced F1- and F2-scores, comparable to those of the neural network, indicating that unsupervised learning can be as effective as supervised learning for this application.

Regarding the other models, LOF model possesses a balanced F1- and F2-score of 87%, with better performances compared to logistic regression and decision tree, and similar to random forest. For what concern the PCA approach,

Table 6 Comparison of the results obtained using unsupervised machine learning algorithms trained using only anomaly-free data

Model	Precision	Recall	F1-score	F2-score	Accuracy
Local outlier factor	78.3%	98.5%	87.4%	93.7%	87.5%
Isolation forest	77.5%	95.2%	85.4%	91.0%	85.7%
Principal component analysis	79.1%	93.3%	85.6%	90.0%	86.2%

Table 7 Comparison of the results obtained using unsupervised machine learning algorithms trained using both normal and anomalous data

Model	Precision	Recall	F1-score	F2-score	Accuracy
Local outlier factor	88.4%	86.7%	87.5%	87.0%	80.4%
Isolation forest	92.9%	91.1%	92.0%	91.5%	87.4%
Principal component analysis	86.0%	84.7%	85.3%	85.0%	76.8%

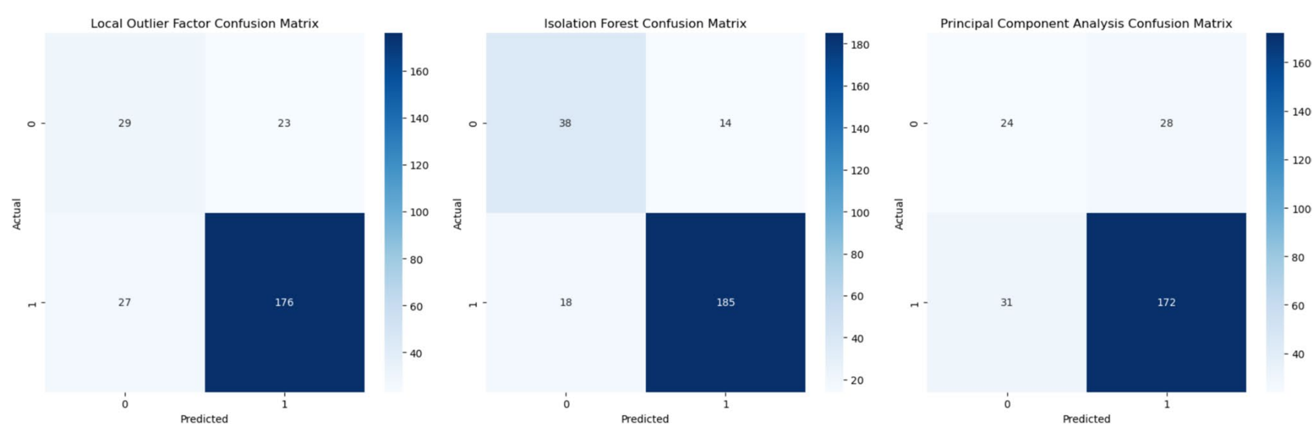


Fig. 17 Confusion matrices showing the results on the test dataset after the models have been trained using an unbalanced mix of good and anomalous deposition data, with the percentage of imbalance known

again, a balanced F1- and F2-score of 85% is reached, with similar but slight worst results of LOF model.

5 Summary and comparison of the results

5.1 Summary

In this study, we conducted a review analysis that identifies a growing trend in the use of unsupervised learning approaches for anomaly detection in WAAM. These approaches are particularly advantageous for handling small and unbalanced datasets, which usually are associated to qualification procedure conducted in real industrial scenario. However, the number of studies in this area is limited, and there is a lack of consistency in the materials, deposition processes, and sensors used, making direct comparisons challenging.

To contribute to this field, in this work, we printed 15 wall structures using Invar 36 alloy with a natural dip transfer process. Our goal was to provide insights into a new material and printing method, as well as to identify optimal parameters for printing this alloy, simulating a process qualification procedure. The optimal set of parameters found is 3.5 m/min of wire feed speed, 21 V of welding voltage, and a welding speed of 4 mm/s.

We collected data on welding current and voltage at sample frequency of 5 kHz to develop an online process monitoring application. The collected data was split into 1-s windows, from which we extracted 83 features in both the time and frequency domains. This approach was chosen because while time-domain information and voltage/current shapes are important, frequency-domain analysis can provide deeper insights into process stability. This concept was described in our previous work, but in this study, it is highlighted through an in-depth analysis of the spectrograms and FFT responses of data associated with various conditions.

At the end of each layer, each 1-s segment (evaluated using the welding speed information) was labelled based on surface appearance, and observations from the welding camera. Out of the 1698 total samples obtained, 20% (344 samples) represented various anomalies, such as excessive spatter, humping, pool shift, arc ignition, porosity, and undercut, reflecting the unbalanced nature of the dataset. Figure 18 summarizes our discussion on selecting the optimal machine learning approach for developing process monitoring applications. The guidance provided is based on key factors including data availability, data collection methodology, and the strictness of certification standards.

Therefore, we applied supervised, semi-supervised, and unsupervised learning methods to the dataset composed by 83 extracted features in both time and frequency domain, with the aim to compare the different approaches based on the collected dataset. In the supervised learning approach, the data was split into an 80–20% training–testing ratio. For semi-supervised learning, only defect-free data was used to train unsupervised models. In the unsupervised approach, the number of anomalies present in the training set was

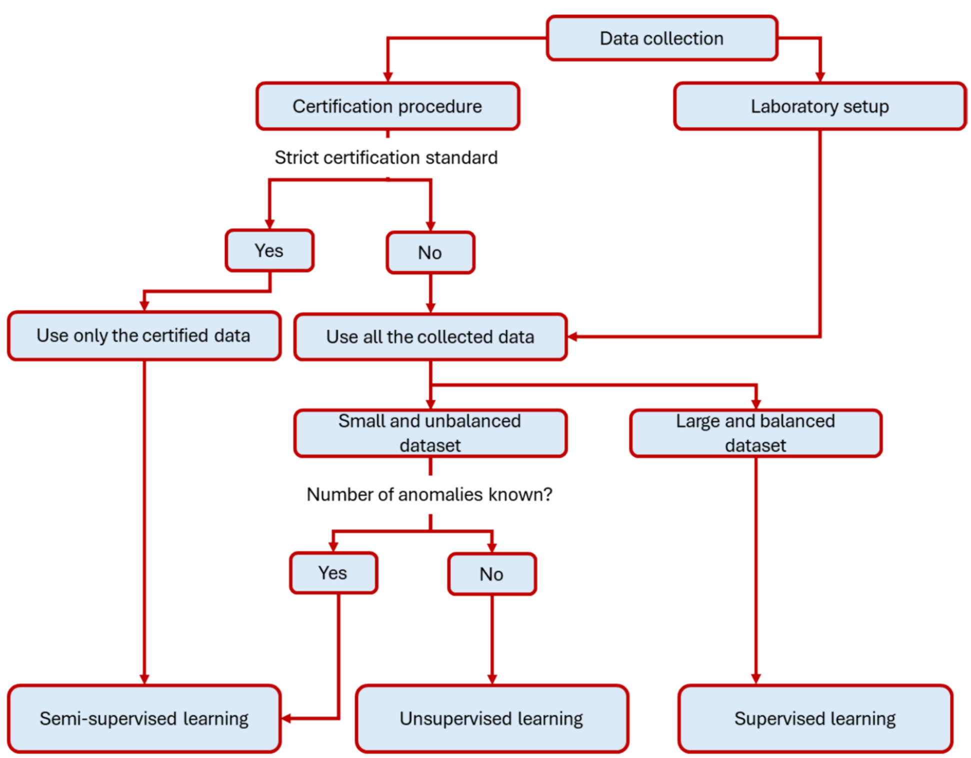


Fig. 18 Guidelines for selecting the optimal machine learning approach in process monitoring applications, considering data availability, data collection methodology, and certification requirements

known (20%), and this information is used to guide the training process, without the need for explicit labels. The results of all approaches were then compared.

5.2 Comparison of the results

In the supervised learning approach, neural networks achieve the highest F1-score of 92.1%, with a recall of 90.4% and precision of 94.0%, indicating strong anomaly detection with minimal false positives. Random forests also perform well, with an F1-score of 89.1% and an accuracy of 95.7%. Decision trees and logistic regression show slightly lower F1-scores, with decision trees having a recall of only 75%, suggesting more missed anomalies. However, observing the F2-score, it is possible observe that all the performances of supervised learning decrease when is given more importance to avoid missing anomalies, which is a typical request in additive manufacturing, since the presence of an anomaly can generate defect in the products. In the semi-supervised approach, where only defect-free data is used to train the models, the LOF shows the best recall at 98.5%, though it sacrifices precision (78.3%), leading to a high number of false alarms. Isolation forest and PCA also show good recall values (95.2% and 93.3%, respectively), but with a slight trade-off in precision. LOF achieves the highest F2-score (93.7%), indicating better performance in terms of detecting anomalies without missing anomalies. However, the reduced capability of generalization reduced the performances of F1-scores compared to supervised approaches. Finally, in the unsupervised setting, where no labels are used but contamination levels are known, isolation forest leads with an F1-score of 92.0% and precision of 92.9%. LOF follows with a precision of 88.4% and recall of 86.7%, while PCA lags behind with an F1-score of 85.3%, indicating weaker overall performance. Generally, in this case, all the models present the best balancing between F1- and F2-scores, indicating that also this approach is valuable and in general is the best to follow if it is important to take advantages from both precision in detect normal cases without losing many anomalies during the deposition process. Therefore, neural networks deliver the best overall performance in the supervised setting. When only defect-free data is available, or labels are not accessible, unsupervised and semi-supervised methods like LOF and isolation forest offer competitive results, with isolation forest showing a solid balance between precision and recall across both semi-supervised and unsupervised paradigms. Despite lower generalization capabilities, unsupervised models can still be effective or can be the only possible solution for anomaly detection in small and unbalanced datasets, though they may suffer from higher false alarm rates, which as discussed is better in AM environment. Nevertheless, the choice of method should align with the available data and the specific goals of the anomaly detection task.

As discussed in the Sect. 2.5 of this work, in highly regulated environments, semi-supervised is more suitable for monitoring deviations with respect to certified parameters, while broader quality assessments in flexible certification contexts can benefit from using the full dataset and therefore form the usage of both supervised and unsupervised learning paradigm. The best results obtained using different ML paradigms are summarized in Table 8.

6 Conclusion

In this study, we investigated the use of machine learning (ML) techniques for anomaly detection in wire arc additive manufacturing (WAAM) of Invar 36 alloy, deposited through the natural dip transfer process. Given the limited prior research on this material in WAAM applications, our goal was to assess its printability and optimize the process parameters through a qualification procedure. To achieve this, we conducted an experimental campaign involving the deposition of 15 wall structures with varying process parameters, providing valuable insights into the behaviour of Invar 36 alloy under different conditions.

After collecting the data, a frequency domain analysis was performed to demonstrate the value of both time-domain and frequency-domain analysis in identifying process anomalies that could lead to product defects. From the 5000 samples corresponding to each 1-s deposition in the welding current and voltage signals, 83 features were extracted using the proposed methodology. These features were then analyzed using a range of ML techniques, including supervised, semi-supervised, and unsupervised learning models, applied to the unbalanced dataset.

The supervised model studied were logistic regression, random forest, decision tree, and neural network, while isolation forest, local outlier factor, and principal component analysis were employed for both semi-supervised and unsupervised learning paradigm.

Table 8 Summary of the results obtained via different ML approaches

Model	Precision	Recall	F1-score	F2-score	Accuracy
Neural network	94.0	90.4	92.1	91.1	96.8
Local outlier factor (unsupervised training)	78.3%	98.5%	87.4%	93.7%	87.5%
Isolation forest (semi-supervised training)	92.9%	91.1%	92.0%	91.5%	87.4%

In the supervised learning approach, neural networks achieve the best F1-score of 92.1%, but the F2-score—which is more important metrics in additive manufacturing—is slight reduced to 91%, given the overfitting which results in some missing anomalies. While this trend is experienced for all supervised learning models, only neural network presents a balanced result between both F1- and F2-scores.

Regarding semi-supervised methods, local outlier factor excels in F2-score with a value of 93.7%, making them suitable when detecting anomalies is paramount, despite higher false alarm rates. The increased values in F2-scores align with the lower F1-score, which for local outlier factor is reduced to 87.4%. Moreover, in this case for all semi-supervised models, a similar trend is provided due to less ability in generalization of the data.

Finally, in the unsupervised setting, isolation forest demonstrates a strong balance between F1- and F2-scores with values of 92% and 91.5%, making it ideal for scenarios where balanced detection is needed without labels. Nevertheless, the analysis conducted in this study highlights that the choice of the best method depends on the data availability and specific requirements, with unsupervised and semi-supervised approaches proving essential for unbalanced datasets in AM environments. The main findings of this study are summarized as follows:

- Preliminary study on printability of Invar 36 alloy under natural dip transfer process highlights that the best process parameters are 3.5 m/min of wire feed speed, 21 V of welding voltage, contact to workpiece distance of 13 mm, and a welding speed of 4 mm/s.
- A time–frequency analysis has been conducted which highlight how the differences between process states and condition (e.g. stable deposition, excessive spatter, pool shift, and porosity) may be appreciated in both time domain signals shape and frequency response.
- The results reached in process anomaly detection by the supervised learning models of logistic regression, random forest, decision tree, and neural network have been compared with the results obtained with isolation forest, local outlier factor, and principal component analysis used in both semi-supervised and unsupervised learning paradigm.
- Supervised learning models perform better on F1-score rather than F2-score since they are more able to detect normal cases, but they increase the possibility of missing detection. They are the perfect solution when large size and fully labelled datasets are available, with neural network achieving the best results with an F1-score of 92%.
- On other hand, semi-supervised models excel on F2-score, since they are subject to higher false alarm rate, which usually is better in AM monitoring applications. These models are the perfect solution when small size data are available without any labels or with only one-class data, e.g. coming from normal deposition data collected during qualification procedure. The best model in this case was the local outlier factor with an F2-score of 93.7% but F1-score of 87.4%
- Finally, unsupervised learning paradigm results in more balanced F1- and F2-scores. They represent the best solution in the presence of small and unbalanced datasets in which there are available some labels. The best model in this case was the isolation forest with a F2-score of 91.5% and F1-score of 92.0%, with similar results of neural network.

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Data availability Data are available under request to corresponding author.

Declarations

Conflict of interest The authors declare no competing interests.

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